COURSE PLAN

FIRST: BASIC INFORMATION

College	
College	Prince Abdullah bin Ghazi Faculty of Information and Communication Technology
Department	Computer Science
Course	
Course Title	Design & Organization of Embedded Systems
Course Code	30801325
Credit Hours	3
Prerequisite	30801220
Instructor	
Name	Prof. Nijad Al-Najdawi
Office No.	
Tel (Ext)	-
E-mail	n.al-najdawi@bau.edu.jo
Office Hours	2:00-4:00pm

Class Times

Building	Day	Start Time	End Time	Room No.
IT	M, W	9:30 AM	11:00 AM	Lab 503

Text Book	
Title	Exploring Arduino Tools and Techniques for Engineering Wizardry, Jeremy Blum, 1st edition

References

https://www.wiley.com/en-jo/Exploring+Arduino%3A+Tools+and+Techniques+for+Engineering+Wizardry-p-9781118549360

www.arduino.org

SECOND: PROFESSIONAL INFORMATION

COURSE DESCRIPTION

This course will expose students to the field of embedded systems and will provide a knowledge foundation which will enable students to pursue subsequent courses in real-time embedded systems software and computer design. Arduino is an open-source, single-board, microcontroller for building devices that can sense and control objects in the physical world. In this instructor-led, live training, participants will learn how to program the Arduino for real-world usage, such as to control lights, motors and motion detection sensors.

COURSE OBJECTIVES

Students successfully completing this course will be able to:

1. To be able to design, describe, validate and optimize embedded electronic systems in different industrial application areas.
2. Ability to solve theoretical and practical problems with emphasis on electronic circuits, control programs and the software-hardware interface.
3. To demystifies the Arduino microcontroller through hands-on work in the lab creating simple machines with embodied behaviors. The Arduino is a versatile resource for physical projects for students in all disciplines.

COURSE LEARNING OUTCOMES

- After successfully completing this course, students will be able to
- Learn how to prototype embedded systems circuits with a breadboard
- Learn the embedded systems programming language (Arduino and IDE)
- Program basic embedded systems examples
- Prototype circuits and connect them to the embedded systems
- Program the embedded systems microcontroller to make the circuits work
- Connect the embedded systems microcontroller to a serial terminal to understand
- communication and stand-alone use
- Explore the provided example code and online resources for extending knowledge
- about the capabilities of the Arduino microcontroller

COURSE SYLLABUS

Week	Course Topic	Notes
Week 1	Getting Started-Introduction to Embedded Systems	
Week 2	Embedded Systems Analysis	
Week 3	Digital Inputs, Outputs, and Pulse-Width Modulation	
Week 4	Reading Analog Sensors	
Week 5	Using Transistors and Driving Motors	
Week 6	Sounds and Multimedia in Embedded Systems	
Week 7	USB and Serial Communication	
Week 8	Shift Registers	**Midterm Exam**
Week 9	The I2C Bus	
Week 10	The SPI Bus	
Week 11	Interfacing with Liquid Crystal Displays	
Week 12	Wireless Communication with XBee Radios	
Week 13	Hardware and Timer Interrupts	
Week 14	Data Logging with SD Cards	
Week 15	Connecting Embedded Systems to the Internet	
Week 16	IOT Systems	**Final Exam**

COURSE LEARNING RESOURCES

This course will be taught using available resources including: lectures, data show and materials uploaded to the e-learning system

ONLINE RESOURCES

Course presentation slides

- University E-Learning

ASSESSMANT TOOLS

(Write assessment tools that will be used to test students ability to understand the course material and gain the skills and competencies stated in learning outcomes

ASSESSMENT TOOLS	%
Quizzes	10
Mid Exam	40
Final Exam	50
TOTAL MARKS	100

THIRD: COURSE RULES

ATTENDANCE RULES

Attendance and participation are extremely important, and the usual University rules will apply. Attendance will be recorded for each class. Absence of 10% will result in a first written warning. Absence of 15% of the course will result in a second warning. Absence of 20% or more will result in forfeiting the course and the student will not be permitted to attend the final examination. Should a student encounter any special circumstances (i.e. medical or personal), he/she is encouraged to discuss this with the instructor and written proof will be required to delete any absences from his/her attendance records.

GRADING SYSTEM

- Check the score-entering system

REMARKS

Use of Mobile Devices, Laptops, etc. During Class, unexpected noises and movement automatically divert and capture people's attention, which means you are affecting everyone’s learning experience if your cell phone, laptop, etc. makes noise or is visually distracting during class. For this reason, students are required to turn off their mobile devices and close their laptops during class.

Academic Integrity. Students, who copy assignments, allow assignments to be copied, or cheat on tests will fail the assignment or test on the first offense, and may fail the entire course on the second. Plagiarism means using words, ideas, or arguments from another person or source without citation. Cite all sources consulted to any extent (including material from the internet), whether or not assigned and whether or not quoted directly.

COURSE COORDINATOR

Course Coordinator: Prof. Nijad Al-Najdawi **Signature:** **Date:** Wednesday, October 2, 2019	**Department Head: Dr. Omar Zobi** **Signature:** **Date:** Wednesday, October 2, 2019